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Effects of an early childhood educator coaching intervention on preschoolers: The role of classroom age composition☆

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ABSTRACT

Heterogeneity in treatment effects of MyTeachingPartner (MTP), a professional development coaching intervention focused on improving teacher–student interactions, was examined for 1407 4-year-old preschoolers who were enrolled in classrooms that served children between the ages of 3 and 5. On average, there were no consistent impacts of MTP coaching on children's school performance, but there was evidence of moderation in treatment effects as a function of classroom age diversity, defined as the proportion of children who were *not* 4 years of age. MTP coaching improved children's expressive vocabulary, literacy skills, and inhibitory control in classrooms that served primarily 4-year-olds and were less age diverse. These effects were in large part due to MTP causing improvements in teachers' instructional support that in turn was more predictive of children's skills in less age-diverse classrooms. Results also indicated that the nature of age diversity did not matter; a greater number of 3- or 5-year-old classmates equally reduced the benefits of the MTP intervention for 4-year-olds. The sole exception occurred for receptive vocabulary, in which case, MTP was most effective in classrooms with a larger number of older (but not younger) children. Taken together, these results suggest that under the right circumstances, the benefits of professional development that improve early childhood educators' teaching practices can also translate into benefits for students.

1. Introduction

Although preschool programs hold promise in preparing children for kindergarten, there is increased interest in understanding how to strengthen program benefits (Duncan & Magnuson, 2013; Phillips et al., 2017; Yoshikawa et al., 2013). Investment in professional development (PD) for early childhood educators has been part of this effort, including opportunities provided through the Race to the Top Early Learning Challenge program. Even with the frequent argument for additional investment in providing teachers with ongoing PD, and the considerable outlay of funds for this purpose, prior evaluations of such services indicate mixed benefits for children (for a meta-analysis see: Markussen-Brown et al., 2017). The scientific community has been urged to now examine more nuanced questions about improving preschool impacts through PD for teachers, notably the range of conditions that potentially moderate PD program impacts (Sheridan, Edwards, Marvin, & Knoche, 2009).

MyTeachingPartner (MTP; Pianta, Mashburn, Downer, Hamre, & Justice, 2008) is a PD intervention that has been found to improve teachers' classroom practices (Downer et al., 2014; Pianta, Masburn et al., 2008), and in prior studies, these benefits have translated into improvements in children's success both in preschool (Mashburn, Downer, Hamre, Justice & Pianta, 2010) and in the secondary grades (Allen, Pianta, Gregory, Mikami, & Lun, 2011). Newer evidence from larger-scale multi-site evaluations of MTP in pre-K classrooms, however, suggests a lack of impact on child outcomes despite improvements in teachers' classroom behaviors (Pianta et al., 2017). A potential focus for an explanation for these equivocal findings, both in the general PD literature and for MTP specifically, is the nature of the classroom setting, particularly conditions that can elevate demands on the teacher and possibly undermine the benefits derived from PD. That is, although PD may actually improve teachers' knowledge or skills, these impacts may not translate into benefits for children unless certain classroom conditions

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are present, one of which, age diversity, we consider in this investigation.

1.1. Coaching interventions and the effectiveness of MyTeachingPartner

PD coaching interventions provide on-going direct support to teachers to improve their knowledge, skills, or teaching strategies that are then applied in their classroom interactions with students (Sheridan et al., 2009). Unlike coursework or workshop trainings, coaching programs offer individualized inputs to teachers based on cycles of observation, implementation, self-reflection and evaluation, and feedback (Sheridan et al., 2009). The MTP coaching program is a PD coaching intervention that focuses on teachers' interactions with students in their classroom, particularly features of instructional support. Teachers receive individualized and ongoing feedback from a coach about specific interaction behaviors with students, as defined and measured by the Classroom Assessment Scoring System (CLASS; Pianta, La Paro, & Hamre, 2008). The goal of MTP coaching is to improve teachers': (a) skills to observe specific features and elements of their interactions with children and their consequences for children's responses; (b) awareness and knowledge of how these interactions contribute to students' early learning; and (c) reflection on their own motivations and tendencies in these interactions.

The theory of change underlying MTP is that by observing effective teacher–child interactions and receiving individualized feedback related to their own interactions with students, teachers' instructional support skills will improve to provide cognitively challenging, yet appropriate, learning opportunities for children through presentation of higher order concepts, opportunities for extended conversations, and consistent, timely, process-oriented feedback (i.e., path a of Fig. 1). These shifts in interaction in turn, will result in improvements in children's early learning (i.e., path b of Fig. 1). This theory of change underlying the MTP coaching model is based on an extensive body of literature that has demonstrated that teachers' day-to-day interactions with students, especially those geared toward stimulating children's thinking and reasoning, shape children's academic learning and executive functioning (Burchinal, Vandergrift, Pianta & Mashburn, 2010; Hamre & Pianta, 2005; Mashburn et al., 2008). Indeed, developmental theory suggests that children's academic development in particular is conditional on the opportunities adults provide them to express existing skills and scaffold more complex ones (Davis & Miyake, 2004; Skibbe, Behnke, & Justice, 2004; Vygotsky, 1978). Classroom interactions with children that are cognitively stimulating, direct, and intentional lay the groundwork for facilitating children's academic and behavioral development, including their language and literacy skills and executive functioning (Johnson et al., 2016; Mashburn et al., 2008; Weiland, Ulvestad, Sachs, & Yoshikawa, 2013).

Even with these potential benefits of high-quality teacher–child interactions, we know that teachers' stimulation and support of children's early learning (i.e., instructional support) is limited in many preschool classrooms across the country (Burchinal, Zaslow, & Tarullo, 2016; LoCasale-Crouch et al., 2007), which means, at present, that children are less likely to reap the maximum benefit from these early learning environments. Given the general pattern of results illustrating the importance of teacher–child interactions (Burchinal et al., 2010, 2016; Hamre & Pianta, 2005; Johnson et al., 2016; Mashburn et al., 2008), the MTP intervention, which was designed to improve teachers' interactions with children, suggests that these impacts—under the right circumstances—can translate into improvements in student's school success. That is, improvements in teacher–child interactions that result from the MTP coaching intervention can facilitate children's early learning outcomes (i.e., path a × path b of Fig. 1).

Prior evaluations of the MTP coaching intervention (with these same data) document significant improvements in teachers' instructional support interactions, which include supporting children's higher order thinking skills (effect size = 0.59), providing intensive feedback (effect size = 0.51), and using language facilitation strategies (effect size = 0.68; Downer et al., 2014). Despite these documented benefits, the quality of teachers' instructional support, on average, is still relatively low even *after* the MTP intervention (CLASS score of 2.76), and these improvements in instructional support have translated only inconsistently into improvements in children's school performance (Mashburn et al., 2010; Pianta et al., 2017). In fact, an intent to treat analysis with the National Center for Research on Early Childhood Education (NCRECE) Professional Development Study (the same data used as this study) documented small impacts, on average, for children's self-regulation (during the year after treatment only) and children's classroom-level language behaviors, but not for academic skills (Pianta et al., 2017). These authors also found no evidence of heterogeneity of effects when looking at a selected group of program- (auspice and dosage), teacher/classroom- (teacher education and curriculum), and child-level factors (home language and pre-test scores) as moderators (Pianta et al., 2017). These authors, however, did not consider the demands that may stem from the diversity in children's needs at the classroom-level, such as the ages of children or the percentage of minority children in a classroom.

Although the MTP coaching intervention has yielded mixed benefits for preschool-aged children in various evaluations (Mashburn et al., 2010; Pianta et al., 2017), there is no question that the classrooms within which these interventions take place are heterogeneous in nature. This is an important point of consideration given that the evaluations of MTP showing few impacts on children were conducted in nine highly diverse locations in eight states (Pianta et al., 2017), when studies detecting impacts on students were done in state-funded pre-K programs in a single state (Mashburn et al., 2010). Characteristics of the work setting (e.g., school-based pre-K or federally funded programs), age of children served (infants, toddlers, and preschool age), and teacher–child ratios have been repeatedly noted as shaping the experiences of children and their teachers (Clarke-Stewart & Allhusen, 2005) in ways that may influence the need for and impact of PD, especially when taken to scale (Sheridan et al., 2009). The PD literature has often focused on the possible moderating impacts of individual children's attributes (e.g., whether a child is an English language learner) and broader classroom/program characteristics (e.g., auspice and curriculum); however, less attention has been devoted to composition and demands of the classroom (e.g., the proportion of English language learners in a classroom). Diversity in classroom-level demands could prove to be a more powerful moderator of treatment impacts than the characteristics of any one individual child, a possibility confirmed in the first evaluation of MTP showing larger benefits for teachers in classrooms in which all children were below the poverty line (Pianta, Masburn et al., 2008).

1.2. Implications of classroom age diversity for professional development

The current study focuses on classroom age diversity as a potential moderator of MTP coaching effects on children. We focus on this aspect of classroom context for a number of reasons. Primarily, the age composition of classrooms is of research and policy interest given that the mixing of ages provides one avenue for increasing 3-year-olds' access to preschool (Phillips et al., 2017). For example, recent national estimates reveal that roughly 75% of Head Start classrooms across the country serve children of different ages (Ansari, Purtell, & Gershoff, 2016), and older estimates from National Center for Early Development and Learning Multi-State

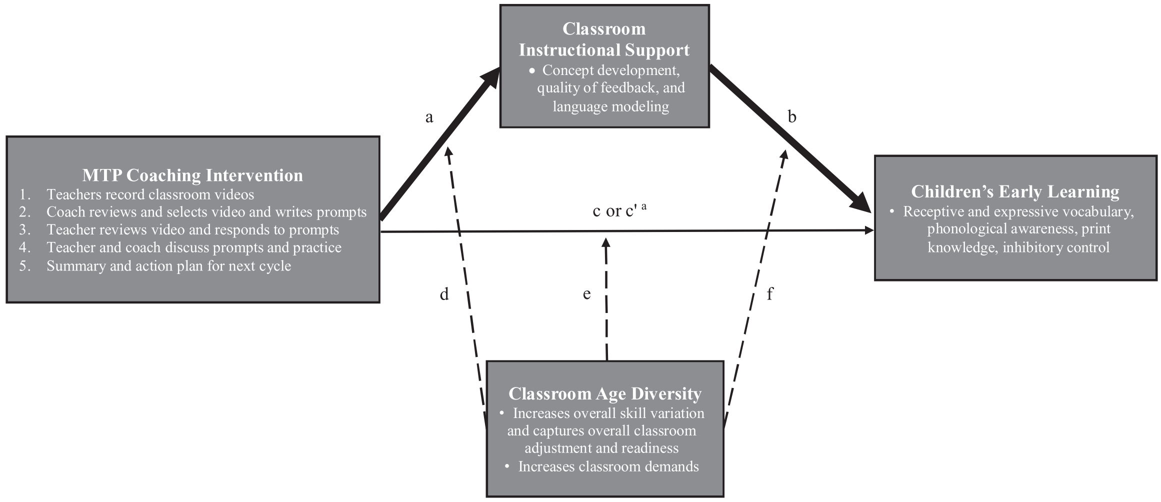

Fig. 1. Conceptual model for the direct, indirect, and conditional effects of the MTP coaching intervention on children's early learning. *Notes.* Dashed lines indicate potential negative effects. Solid lines indicate potential positive effects. Bolded lines indicate indirect effects. [a]Path c corresponds to a model in which instructional support is *not* included, whereas path c' corresponds to a model in which instructional support is specified.

Study (Bryant et al., 2002) reveal that over a quarter of state-funded pre-K programs served children of different ages in the same classroom (authors calculations). And although age composition data were not available in the original MTP evaluation (Mashburn et al., 2010; Pianta, Masburn et al., 2008), estimates by the National Institute for Early Education Research (Barnett et al., 2004, 2005) indicate that Head Start and pre-K programs (at the time) only served 4% of 3-year-olds in the participating state. In contrast, estimates from the eight states participating in the NCRECE study reveal that these programs served 12% of 3-year-olds (Barnett, Epstein, Friedman, Sansanelli, & Hustedt, 2009; Barnett et al., 2010), suggesting that there is a sizable difference in the ages of children served across the two MTP evaluations.

From a theoretical perspective, support for classroom age diversity is rooted in both sociocultural theory (Vygotsky, 1978) and social-learning theory (Bandura, 1986), which focus on the role of peers in children's learning and development. These models suggest that interacting with both older and younger children can shape children's school success, including children's language and literacy development and executive functioning, as it provides opportunities for modeling more advanced behavior and scaffolding learning for younger peers. And even though these developmental theories have been the bedrock for much of the peer-effects literature (e.g., Henry & Rickman, 2007; Justice, Petscher, Schatschneider, & Mashburn, 2011; Mashburn, Justice, Downer, & Pianta, 2009), whether these theories work in practice has remained contested given the everyday challenges that age diversity presents.

For example, because age can be a marker for a wide range of skill levels and abilities (Mason & Burns, 1996), this source of diversity may add to the demands on a teacher to adjust instruction in the interest of providing individualized support for students in their classroom. These issues are particularly pertinent in the context of pre-K classrooms because: (a) there are large developmental differences between 3-, 4-, and 5-year-olds and (b) the strict age cutoffs for kindergarten entry limit age diversity in other grade levels. In support of these very points, there is some empirical evidence to suggest that pre-K teachers demonstrate less optimal instructional and emotional support, and classroom organization in classrooms of greater age diversity (Ansari & Pianta, 2018). That is, classrooms in which children vary widely by age can be notably more challenging for a teacher in terms of fostering children's self-regulation, addressing needs for care and safety, and responding to children's levels of autonomy or peer competence.

Finally, a number of studies have also considered the implications of classroom age diversity for children's early learning. Although the evidence is far more mixed when looking at children's early academic and social-behavioral development, with some scholars documenting positive effects (Blasco, Bailey, & Burchinal, 1993; Goldman, 1981; Guo, Tompkins, Justice, & Petscher, 2014), and others documenting null or negative associations (Ansari et al., 2016; Bell, Greenfield, & Bulotsky-Shearer, 2013; Moller, Forbes, Jones, & Hightower, 2008; Urberg & Kaplan, 1986; Winsler et al., 2002), there is a growing recognition that age diversity is a relevant feature of the classroom ecology. That is, higher skilled (and possibly older) peers can foster children's early learning and development by modeling more complex language and behaviors (Henry & Rickman 2007; Mashburn et al., 2009; Winsler et al., 2002), especially in classrooms of higher quality (Guo et al., 2014). At the same time, it is often hypothesized that classroom age diversity results in a more demanding classroom in which to teach, and therefore, is a factor to consider when evaluating the impacts of early intervention programs (Ansari & Pianta, 2018; Mason & Burns, 1996). Supporting these general notions, two large-scale studies of publicly funded preschool programs in the United States found that 4-year-olds enrolled in classrooms with a greater number of younger peers demonstrated fewer gains in areas of early language, literacy, and math (Ansari et al., 2016; Moller et al., 2008).

The question, therefore, is whether the demands imposed by the age diversity of classrooms nullify the possible benefits derived from coaching that improves teachers' classroom practices. In other words, although PD can (and does) increase teachers' skills for promoting children's early learning and development (Downer et al., 2014; Hamre et al., 2010; Hamre et al., 2012; Markussen-Brown et al., 2017), these skills might only have an aggregate impact for children under circumstances related to the age levels of the children in the classroom.

Because we know that classroom age diversity increases the demands experienced by teachers (Ansari & Pianta, 2018), it is plausible that age diversity reduces the benefits *teachers derive* from the MTP coaching intervention (i.e., path d moderates path a of Fig. 1). Considering that prior evaluations of MTP with these same data have yielded aggregate (and sizable) impacts across classrooms of different ages (Downer et al., 2014), this possibility may not apply here. If age diversity were minimizing the benefits teachers derived from MTP, it is likely that the aggregate impacts of the intervention would be considerably smaller. A second possible manner in which age diversity conditions PD impacts is that age diversity reduces the benefits *children derive* from the (improved) classroom environment (i.e., path f moderates path b of Fig. 1). Thus, age diversity may be one reason why recent studies of preschool programs have documented only small associations between classroom quality and children's school success (e.g., Keys et al., 2013; McCoy, Connors, Morris, Yoshikawa, & Friedman-Krauss, 2015; Weiland et al., 2013).

In terms of the MTP coaching program specifically, although this intervention does attempt to focus on supporting teacher's ability to provide higher quality instruction and has been used for preschool-aged children (Pianta, Masburn et al., 2008), the lack of aggregate impacts might stem from challenges with individualization or with the overall low level of instructional support even after MTP. From an individualization perspective, the fact that although teachers are learning to better attend to children's cues, these new skills may be overwhelmed in an age-diverse classroom in which children demonstrate a wider range of needs. That is, in an age-diverse classroom, overall higher quality instructional support on the part of the teacher may not assure that all children are receiving instructional content appropriate to their individual needs (Burchinal, 2017). For this reason, the MTP intervention may produce improvements in teachers' provision of higher quality instructional support, but these improvements may be less beneficial for children in age-diverse classrooms. Second, recall that even though the MTP program improves teachers' instructional support, teachers still score quite low on average on this dimension, and the level of instructional support (even after the intervention) may reflect a lack of behaviors most relevant to the needs of all children in an age-diverse classroom.

Despite the fact that these two different points at which age diversity might alter the benefits children derive from the MTP intervention could have varying implications for the meaning of classroom age diversity, either pathway results in moderated mediation. Put another way, if path a of the conceptual model that links the impacts of the MTP intervention with teachers' instructional support is altered by age diversity, then it is possible that the problem is not age diversity per se, but that MTP does not improve teachers' skills to the level needed to address children's needs in age-diverse classrooms. Conversely, if pathway b of the model, linking post-MTP classroom instructional quality to children's early learning, is modified by age diversity, then it may be that the MTP intervention is not targeting the specific teacher skills necessary to individualize their instruction in age diverse preschool settings. But, as noted above, if either pathway of the conceptual model is altered by age diversity of classrooms, the result is the same: children benefit less from the MTP coaching intervention (i.e., path e moderates path c of Fig. 1).

1.3. The current study

As part of the current investigation, we revisit the work of Pianta et al. (2017) who studied the impacts of the MTP coaching intervention for children's early language, literacy, and executive functioning, but found limited evidence to suggest that the intervention impacts for teachers translated into benefits for 4-year-old children. As part of the current study, however, we attempt to advance the field through a study of moderated mediation of a PD intervention, which to our knowledge has been rarely investigated or detected despite numerous calls to examine the varying conditions under which PD is or is not effective (Sheridan et al., 2009). Specifically, we examine how the putative impacts of MTP coaching on teachers' interactions with students in turn translate into benefits for students under conditions of age diversity. To do so, we use data from 1407 preschool-aged children whose teachers participated in the NCRECE Teacher Professional Development Study to address the following questions:

1. To what extent are the impacts of the MTP coaching intervention attenuated in preschool classrooms with greater age diversity as compared with preschool classrooms with less age diversity?
2. To what degree are these impacts (or lack thereof) of the MTP coaching intervention accounted for by changes in teachers' instructional support, which is one of the main targets of the MTP program?

When taken together, we hypothesize that the benefits of the MTP coaching intervention will be larger for children when age diversity is lower. Although we expected that these findings would be attributed—at least in part—to the main target of the MTP intervention (i.e., teachers' instructional support), we left this second question as largely exploratory given the competing hypotheses outlined above (i.e., age diversity reduces the benefits teachers derive vs. age diversity reduces the benefits children derive from the classroom environment). In addressing these possibilities, the results of this investigation can begin to provide important insight into the broader contextual conditions under which the impacts of coaching interventions are optimized and, in turn, have benefits for students in the classroom, which is increasingly necessary given the current landscape and expansion of early childhood education in the United States.

2. Method

2.1. Recruitment and participants

Two cohorts of teachers (Cohort 1: 2008–2010 and Cohort 2: 2009–2011) were recruited from Head Start programs and other large community-based preschool programs in urban locations, including: New York City, NY; Hartford, CT; Chicago, IL; Stockton, CA; Dayton, OH; Columbus, OH; Memphis, TN; Charlotte, NC; and Providence, RI. As part of the recruitment, teachers were eligible for participation if: (1) they were the lead teacher in a publicly funded classroom; (2) delivered classroom instruction primarily in English; and (3) they had access to high-speed Internet at the program. This recruitment effort resulted in 401 preschool teachers who were randomly assigned to receive PD (i.e., treatment condition) or to go about their business as usual (i.e., control condition). These teachers were randomly assigned to the treatment or control conditions within each urban location/site (i.e., New York City, NY; Hartford, CT, etc.), which meant that in some cases, teachers in the same center could (and in some cases, did) participate in different conditions. All teachers received a $100 gift card for each phase of participation (for a maximum of $300).

For the purposes of the current study, we focus on the 325 teachers who had students from their classroom participate in data collection. These teachers were 42 years of age (SD = 10.79), averaged almost 16 years of education (SD = 1.62), and had roughly 14 years of teaching experience (SD = 9.28). The majority of the participating teachers were either African American (47%) or White (33%), with a smaller number identifying as Latino (13%) or Asian/other (7%). Importantly, and as can be seen in Table 1, randomization was largely achieved across the treatment and control conditions when looking at teacher and classroom characteristics.

To recruit children, participating teachers (who were already randomized to either the treatment or control condition) distributed recruitment packets at the beginning of the year to parents of all children in their classroom. Completed packets were collected from parents, which were used to randomly select and assess approximately four children per classroom, with preference given

Table 1
Sample descriptives separated by treatment and control conditions.

	MTP treatment group	Control group	Sig. Diff.
Teacher and classroom level variables			
Female	0.95	0.97	
Age	42.68 (11.20)	42.10 (10.37)	
Black	0.44	0.50	
White	0.34	0.31	
Latino	0.16	0.10	
Asian	0.06	0.09	
Years of education	15.76 (1.70)	15.96 (1.52)	
Years of experience	14.56 (9.55)	14.52 (9.20)	
Has received a CDA	0.26	0.27	
Extraversion	3.65 (0.41)	3.66 (0.41)	
Agreeableness	4.15 (0.42)	4.10 (0.39)	
Conscientiousness	4.08 (0.44)	4.11 (0.47)	
Literacy curriculum	0.24	0.23	
Speak Spanish in the classroom	0.42	0.46	
Instructional support strength	3.91 (0.68)	4.04 (0.66)	
Emotional support strength	4.21 (0.58)	4.33 (0.58)	
Organizational support strength	3.86 (0.78)	4.05 (0.70)	*
Participated in year 1 intervention	0.39	0.38	
Participated in year 1 control	0.40	0.44	
Did not participate in year 1	0.21	0.18	
Classroom instructional support			
Pre MTP intervention	2.09 (0.45)	2.04 (0.47)	
Post MTP intervention	2.76 (0.68)	2.35 (0.51)	***
Works in Head Start	0.54	0.61	
Works in a public school	0.38	0.36	
Class size	17.71 (3.50)	18.03 (3.35)	
Classroom racial/ethnic diversity	0.38 (0.22)	0.38 (0.23)	
% of children 2 or 3 years of age	20.60 (24.39)	19.65 (20.80)	
% of children 4 years of age	53.13 (22.53)	58.61 (22.37)	*
% of children 5 years of age	26.27 (21.73)	21.74 (21.08)	
% of girls in classroom	48.06 (12.38)	48.96 (10.53)	
% of children with disability	7.52 (9.87)	7.07 (9.40)	
% of children with limited English	13.24 (17.40)	16.19 (23.17)	
Class income-to-needs ratio	1.11 (0.80)	1.05 (0.59)	
Teacher level sample size	168	157	
Child level variables			
Age	4.16 (0.48)	4.18 (0.45)	
Male	0.50	0.49	
Black	0.45	0.49	
Latino	0.37	0.31	*
White	0.13	0.10	
Asian	0.06	0.09	*
English home language	0.86	0.88	
Household income/1000	24.32 (22.77)	23.10 (19.49)	
Mothers' years of education	12.73 (2.06)	12.74 (2.02)	
Father lives at home	0.46	0.47	
Beginning of year skills			
Receptive vocabulary	85.83 (16.81)	86.25 (16.90)	
Expressive vocabulary	96.09 (17.13)	96.03 (16.46)	
Phonological awareness	90.42 (13.99)	89.45 (13.95)	
Print knowledge	96.01 (14.57)	96.02 (14.67)	
Inhibitory control	0.46 (0.32)	0.46 (0.33)	
End of year skills			
Receptive vocabulary	90.38 (16.19)	89.59 (15.92)	
Expressive vocabulary	96.82 (13.98)	96.88 (13.08)	
Phonological awareness	93.46 (15.07)	92.48 (14.98)	
Print knowledge	102.23 (14.48)	101.63 (14.70)	
Inhibitory control	0.64 (0.32)	0.64 (0.33)	
Child level sample size	736	671	

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

to 4-year-olds ($M = 4.17$, $SD = 0.47$; 11% of children were 42 months of age or younger and 28% of children were 54 months of age or older). Children were considered eligible for the study if they had no IEP at the beginning of the year and spoke English (87%) or Spanish (13%). These sampling procedures resulted in 1407 participating children, 52% of whom had teachers who received MTP coaching. The majority of participating children were enrolled in Head Start (58%), with a smaller portion attending public school pre-K (37%; not mutually exclusive). Although there was an even distribution of boys (49%) and girls (51%) in the study sample, the largest share of children were Black (47%) or Latino (34%) with a smaller number of children coming from a White (11%) or Asian/other (8%) household. For other child- and family-level demographic characteristics, see Table 1.

MyTeachingPartner (MTP) coaching intervention. The MTP coaching intervention, which is described in more detail in the introduction, involved observation-based analysis and web-mediated feedback for teachers. All participating teachers received video cameras to record their classroom activities and, over the course of a school year, teachers videotaped their implementation of an instructional activity (e.g., literacy or language) and sent the recorded activity to their coach on a biweekly basis. The coach edited the tape into three segments that focused on specific dimensions of teacher–child interactions, as they were evident in teachers' interactions with various children in the tape (e.g., concept development, behavior management, etc.). These clips targeted: (a) identification of effective interactions; (b) analysis and identification of alternative approaches to interaction that could prove more beneficial for engaging the particular child or children; and (c) improving instructional support behaviors. The segments include feedback and questions that were shared with teachers who then viewed the tapes and responded to the consultant's questions and prompts. After the teachers viewed the videos and read/responded to the coach's questions and prompts, the teacher and coach participated in a videoconference where they discussed the videos and the strengths and weaknesses of their interactions with students in their classroom. On average, teachers participated in 10 cycles of coaching throughout the school year (M = 10.23, SD = 4.06, range = 1–21). The average number of cycles attended by the teachers in this study is aligned with the target number of cycles required for successful implementation of MTP (8–12 cycles; Allen et al., 2011; Downer et al., 2011), and over 70% of teachers in this study sample received at least the minimum recommended dosage of MTP coaching (eight or more cycles). The first three cycles of the MTP coaching program focused on the Emotional Support domain of the CLASS, followed by two cycles on Classroom Organization, and the remainder, and majority, of the sessions focused on the Instructional Support domain.

With regard to the coaches, 12 individuals with expertise in early childhood education and who had a graduate school education were hired as part of the study within each urban location. Although coaches had a wide range of background experiences, the coaching process was highly standardized and manualized across sites and included pre-training reading assignments, CLASS reliability training, and role-play modeling of coaching. Across the school year, coaches participated in refreshers that provided them with further prompt writing and conferencing practice (for more information on coaches, see Pianta et al., 2017). Each of the 12 coaches received weekly reviews and participated in biweekly conference calls with the entire team of coaches and NCRECE staff. All facilitators also had access to logs of teachers' participation in the coaching program and, thus, monitored coaches on a weekly basis. In addition, a member of the NCRECE staff completed a 23-item fidelity checklist several times throughout the year for each coach, with total scores representing the percentage of the intervention components completed by a coach. Based on these ratings by the NCRECE staff, coaches averaged a fidelity rate of 94% (range 91–97%; LoCasale-Crouch et al., 2016). Teachers also rated the quality of their experiences with their coaches several times a year with 10 survey items based on a 4-point scale, and on average, rated their coaches quite highly (M = 3.50, SD = 0.41; Pianta et al., 2014).

2.2. Measures

Classroom age composition. At the start of the school year, teachers reported on the number of children in their classroom—not just study children—and how many were 2, 3, 4, or 5 years of age. Based on these reports, we divided the number of children in each category by the class size to generate the proportion of 2-, 3-, 4-, and 5-year-olds. Given the small number of 2-year-olds (less than 1% of children in the classroom), we combined children who were 2 and 3 years of age into one category. It is important to note that in reporting on *all* children's ages in the classroom, our focal measure of classroom age composition included the target child's age. As a precaution, we estimated a series of models that dropped the focal child's age from our classroom age composition indicators. Reflecting the fact that we had access to the ages of *all* children in each classroom, which on average included 18 students (as opposed to 4–5 children per classroom in much of the peer effects literature; Justice et al., 2011; Mashburn et al., 2009), all results were the same as those reported below (results available from authors). And given the age distribution of classrooms (i.e., they were largely serving 4-year-olds), those that enrolled a greater share of 3- or 5-year-olds were considered to be more age diverse, whereas classrooms that served more 4-year-olds were considered to be less age diverse. For standardized mean differences in the proportion of 4-year-olds (vs. 3- and 5-year-olds) as a function of the covariates, see Appendix Table 1.

Instructional support. During the beginning and end of the preschool year, teachers were observed and rated on the CLASS (Pianta, La Paro et al., 2008) by trained observers. The CLASS, which is based on a 7-point Likert scale (1–2 = low to 6–7 = high), was used to measure three different domains of teacher–child interactions, namely: (a) Emotional Support (i.e., positive and negative climate, teacher sensitivity, and regard for student perspectives), (b) Classroom Organization (i.e., behavior management, productivity, and instructional learning formats), (c) and Instructional Support (i.e., concept development, quality of feedback, and language modeling). Prior validation studies with the National Center for Early Development and Learning Multi-State Pre-K Study (Bryant et al., 2002) have shown that inter-rater agreement for these CLASS scales have ranged from 0.72 to 0.89, and factor analyses of over 4000 classrooms reveal a similar three-factor solution with alphas ranging from 0.81 to 0.89 (Hamre, Pianta, Mashburn, & Downer, 2007). For the purposes of this investigation, we focus on the Instructional Support domain of the CLASS, because prior studies with these data have found that this domain of teacher–child interactions improves as a result of MTP coaching (Downer et al., 2014). For descriptives, see Table 1.

Child outcomes. Three dimensions of children's early learning and development were directly assessed in the fall and spring of the school year and were primarily selected because they have been found to be influenced by the quality of teacher–child interactions (e.g., Johnson, Markowitz, Hill, & Phillips, 2016; Mashburn et al., 2008; Weiland et al., 2013). The measures discussed below were also selected because they were objective assessments of children's early learning and development rather than based on teacher report, which may be biased, especially in the context of classrooms that enroll children of different ages (see Ansari et al., 2016).

First, children's language skills were assessed with two measures that captured children's *receptive* and *expressive vocabulary*. Specifically, children's receptive vocabulary was assessed with the then available third edition of the Peabody Picture Vocabulary Test (α = 0.95; Dunn & Dunn, 1997), which required that assessors ask children to point to one of four pictures that best illustrated the meaning of a word that was said aloud by the assessor. To assess children's *expressive vocabulary*, each child was individually administered the Picture Vocabulary subtest of Woodcock–Johnson III Psychoeducational Battery (α = 0.81; Woodcock, McGrew, & Mather, 2001), which required that children name objects depicted in a series of pictures. For our purposes, we used the age-standardized scores for these assessments. Next, the Test of Preschool Early Literacy (Lonigan, Wagner, Torgesen, & Rashotte, 2007) was used to measure two different dimensions of children's literacy skills: *phonological awareness* and *print knowledge*, both of which have also demonstrated good inter-

Table 2
The effects of MTP coaching and coaching × age composition interactions for children's early learning.

	Receptive language	Expressive language	Phonological awareness	Print knowledge	Inhibitory control	Average of all child outcomes
Main effect model						
MTP coaching intervention	0.07[†]	−0.02	0.01	0.06	0.02	0.03
	(0.04)	(0.04)	(0.05)	(0.04)	(0.05)	(0.04)
Interaction models[a]						
Coaching × proportion of 4-year-olds	0.00	0.11[**]	0.09[*]	0.11[*]	0.17[***]	0.12[***]
(proportion of 3- and 5-year-olds)	(0.04)	(0.04)	(0.05)	(0.05)	(0.05)	(0.04)
Coaching × proportion of 3-year-olds	−0.05	−0.12[**]	−0.09	−0.12[*]	−0.15[*]	−0.14[***]
(proportion of 4-year-olds)	(0.04)	(0.04)	(0.06)	(0.06)	(0.06)	(0.04)
Coaching × proportion of 5-year-olds	0.05	−0.10[*]	−0.10[†]	−0.10[*]	−0.17[**]	−0.09[*]
(proportion of 4-year-olds)	(0.04)	(0.04)	(0.06)	(0.05)	(0.05)	(0.04)
Coaching × proportion of 5-year-olds	0.09[*]	0.01	−0.01	0.01	−0.03	0.04
(proportion of 3-year-olds)	(0.04)	(0.04)	(0.06)	(0.05)	(0.05)	(0.04)

[a] The text in brackets corresponds to the reference category, which was rotated to get all pairwise comparisons. All continuous variables have been standardized to have a mean of zero and standard deviation of one and, thus, all parameter estimates correspond to effect sizes. Estimates in brackets are standard errors.
[†] $p < 0.10$.
[*] $p < 0.05$.
[**] $p < 0.01$.
[***] $p < 0.001$.

nal consistency (α's = 0.78–.89). Children's phonological awareness measured children's word elision and blending, whereas the print knowledge subtest measured children's knowledge of the alphabet and conventions of the written language. For these two assessments, only raw scores were available and, thus, we used raw scores for children's literacy skills. Finally, children's *inhibitory control* was assessed with an adapted version of the pencil-tapping task, which required that children tap once when the assessor tapped twice and vice versa (adapted from Diamond & Taylor, 1996), which is a measure of inhibitory control that has demonstrated adequate reliability across trials (α's > 0.80; Bierman, Nix, Greenberg, Blair, & Domitrovich, 2008; Blair & Razza, 2007). For our purposes, we used the percent of correct responses. In addition to looking at each individual outcome, we also standardized all outcome scores within our study sample to have a mean of 0 and standard deviation of 1 so that each measure was based on the same distribution. We used these standardized measures to create an overall composite of children's early learning to gauge the average impact of the intervention across domains of children's development (α's = 0.78–.79). These scores were then re-standardized to have a mean of 0 and standard deviation of 1.

It is important to note that the above batteries were for the most part administered to children in English, because teaching in the study classrooms was done predominantly in the English language. Accordingly, children were administered the Picture Vocabulary and the Test of Preschool Early Literacy in English. Children whose primary language was English were also administered the Pencil Tap task and the Woodcock–Johnson in English, whereas bilingual children whose primary language was Spanish were administered these assessments in Spanish by a bilingual assessor. We recognize the complex issues surrounding home language, language of instruction, and language of testing identified in the literature. Because the intention of the measures used in this study was to measure progress in English-language skills taught in these classrooms (rather than total language or conceptual language skills), we opted to include all children in the analyses and adjust for home language. As a precaution, however, we also estimated models excluding these children for whom English-language assessments may have been inappropriate; all results were the same as those reported below (results available from authors).

Covariates. Despite the fact that balance was largely achieved at the level of randomization, we opted to control for a rich set of covariates because teachers and children were not randomly assigned to classrooms as a function of age composition (i.e., our moderator). Thus, the covariates in our models included teacher- and classroom-level characteristics in addition to a series of child and family demographics. Statistically speaking, the inclusion of these covariates allows us to isolate heterogeneity in treatment effects that result from age composition from other factors that might covary with classrooms that serve different age children. That is, controlling for these indicators reduces the possibility of spurious associations and ensures that we are identifying meaningful associations between our variables of interest.

To begin, teacher-level characteristics included the following: teachers' gender, teachers' age, teachers' race/ethnicity (White, Black, Latino, and Asian), teachers' years of education, teachers' years of experience, teachers' receipt of a CDA (0 = no, 1 = yes), teachers' use of Spanish in the classroom (0 = no, 1 = yes), teachers' use of a literacy curriculum (0 = no, 1 = yes), whether teachers worked in a Head Start or public school program, and teachers' agreeableness, extraversion, and consciousness as measured by the NEO Personality Inventory (Costa & McCrae, 1985). We also controlled for teacher self-reports of their strengths and challenges, which consisted of 10 items that required teachers to rate how well they interacted with the children in their classroom at the start of the school year. These items were rated on a response scale from 1 (area for much growth) to 5 (area of great strength) and captured strengths and challenges in instructional support (4 items, α = 0.79), emotional support (3 items, α = 0.64), and classroom organization (3 items, α = 0.77).

Next, at the classroom level, we controlled for the following: class size, percent of girls in the classroom, percent of children in the classroom with special needs, percent of children in the classroom with limited English skills, racial/ethnic diversity of the classroom (using Simpson's Diversity Index, 1949), and the classroom income-to-needs ratio. Finally, at the child and family level, all models adjusted for the following: child's age, child's race/ethnicity (White, Black, Latino, and Asian), child's gender, household income, mothers' years of education, whether the child's father lived at home, child's home language (Spanish or English), and lagged dependent variables as captured by children's school entry skills. Thus, all models considered the extent to which the MTP coaching intervention resulted in *gains* in children's early learning and development. All models also adjusted for site fixed effects (i.e., New York City, NY; Hartford, CT; Chicago, IL; Stockton, CA; Dayton, OH; Columbus, OH; Memphis, TN; Charlotte, NC; Providence, RI) to account for any variation across study locations.

2.3. Analytic strategy

We estimated average treatment effects within a regression framework in the *Mplus* program (version 8; Muthén and Muthén, 1998–2013). To account for dependence in child outcomes, we clustered models at the classroom level and missing data, which ranged from roughly 0 to 17%, were addressed with full information maximum likelihood estimation (Schafer & Graham, 2002). For our moderation analyses of child outcomes, we interacted our focal predictor (treatment) with our indicators of classroom age composition (i.e., the proportion of 3-, 4-, and 5-year-olds). For these analyses, we rotated the age composition referent to provide each pairwise comparison. If there was evidence for moderation, we then plotted the interactions by calculating the predicted outcome scores for different combinations of the MTP coaching intervention and classroom age composition and examined the regions of significance (see Aiken & West, 1991).

These analyses were followed up with a series of multi-group path models that considered the extent to which children benefited from MTP coaching as a result of improvements in teachers' instructional support and whether these effect were moderated by classroom age composition (i.e., models of moderated mediation). To formally test for moderated mediation, we used the INDIRECT command in the *Mplus* program coupled with a Wald's test to compare coefficients across the different strata of age composition (i.e., classrooms with less age diversity as compared with classrooms with greater age diversity).

3. Results

On average, 4-year-olds who participated in the NCRECE Professional Development Study attended classrooms in which roughly 20% of their peers were 3 years of age (SD = 23%), 56% of their peers were 4 years of age (SD = 22%), and 24% of their peers were 5 years of age (SD = 23%). Thus, the study children attended preschool classrooms in which 44% of their classmates were of a different age; but as can be seen by the standard deviation of these classroom level indicators, there was also a great deal of variation in the age composition of classrooms. Descriptive analyses also revealed that both children in the MTP treatment and control conditions, on average, demonstrated gains in all domains of early learning and development across the preschool year (see Table 1). Nonetheless, similar to the intent to treat analysis by Pianta et al. (2017), we found no consistent impacts of the MTP coaching intervention for the direct assessments of children's early language and literacy development or their executive functioning, both in the bivariate (see Table 1) and multivariate models (see Table 2). In fact, across all five outcomes of interest, the average impact of the MTP coaching intervention was not statistically significant and corresponded to approximately 3% of a standard deviation, suggesting that both children in the treatment and control conditions demonstrated comparable gains across the school year (i.e., there was no support for path c of Fig. 1).

Heterogeneity in MTP coaching effects. Despite the lack of aggregate impacts of MTP coaching that translate into benefits for children, results from our moderation analyses indicated that the PD intervention consistently was more effective under certain conditions related to the age composition of the classroom (i.e., path e of Fig. 1 moderates path c). Specifically, and as can be seen in the interaction plot presented in Fig. 2, significant treatment × age composition interactions revealed that the coaching intervention had no detectable benefits for children for four of the five outcomes of interest when 4-year-old children were enrolled in classrooms with greater age diversity. For example, in classrooms with greater age diversity where 50% of children were either 3 or 5 years of age, the intervention had no detectable benefits for children's early learning (ES = 0.00, ns). In contrast, for classrooms in which there was less age diversity (65–80% of peers were 4 years of age), the coaching intervention resulted in greater improvements in 4-year-old's language, literacy, and executive functioning (ES = 0.10–0.15, $ps < 0.05$; for regions of significance, see Fig. 2). Results also suggested that the source of diversity did not matter; that is, having a larger number of 3- or 5-year-old classmates equally reduced the benefits of the coaching intervention. The sole exception to this general pattern of findings was with respect to children's receptive vocabulary. In this one instance, we found that when children had a greater number of older classmates (i.e., 5-year-olds), the MTP coaching intervention resulted in greater improvements in children's receptive vocabulary.

Indirect effects of MTP coaching. Having established that the MTP coaching intervention resulted in improvements in children's early learning in classrooms that were less age diverse and served largely 4-year-olds, we next considered potential explanations as to why this might be the case, with a specific focus on changes in teachers' instructional support, which prior studies with these data have found improves because of MTP coaching (Downer et al., 2014). For simplicity, our next set of analyses focuses on the average child outcomes across the five domains of early learning among children in the top (73%+ 4-year-olds, M = 87% 4-year-old classmates, less age diversity) and bottom (<42% 4-year-olds, M = 29% 4-year-old classmates, greater age diversity) quartiles of the age composition distribution, which provided adequate power at the child-level to detect effect sizes of approximately 0.15 or greater and group differences of roughly 0.20 or greater. It is important to note that (a) we examine a composite of children's early learning because, as discussed above, this composite largely captured our study findings; and (b) we estimated multi-group models across the top and bottom quartiles of the age composition distribution because this allowed us to use the INDIRECT command in the *Mplus* program to test for moderated mediation.

As can be seen in Table 3, results from this effort revealed that the MTP coaching intervention resulted in statistically comparable improvements in teachers' instructional support in classrooms of greater age diversity (ES = 0.73, $p < 0.001$) and classrooms of less age diversity (ES = 0.60, $p < 0.001$). That is, the age diversity of classrooms did *not* alter the benefits teachers derived from the intervention, which corresponds to path d of our conceptual model. Teachers' instructional support, however, only resulted in improvements in children's early learning in classrooms of less age diversity, which served largely 4-year-olds (i.e., path f of our conceptual model). In fact, the benefits of instructional quality for children's early learning were approximately 35% of a standard deviation greater in classrooms that served more 4-year-olds

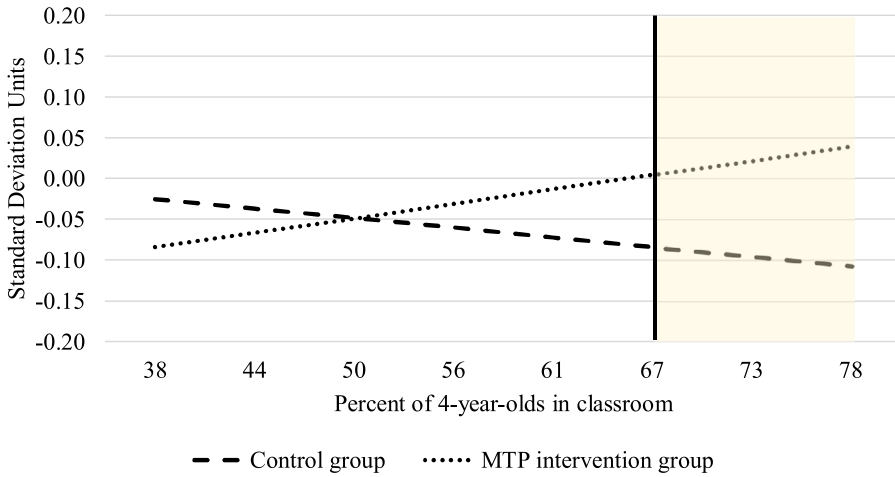

Fig. 2. The conditional impacts of coaching for children's early learning (average of all outcomes) as a function of the distribution of 4-year-olds. *Note*. The average child was enrolled in classrooms where roughly 56% of his or her peers were 4 years of age. All other contrasts correspond to a +/−.25SD change in the percent of 4 year olds. Shaded areas correspond to the regions of significance.

Table 3
Direct and indirect effects of the MTP coaching intervention across classrooms of different levels of age diversity.

Predictor		Mediator		Outcome	Overall	Greater age diversity	Less age diversity
Direct effects							
MTP coaching intervention	→	Instructional support			0.78*** (0.10)	0.73^{***a} (0.17)	0.60^{***a} (0.17)
		Instructional support	→	Average of all child outcomes	0.04 (0.03)	−0.06^{a} (0.08)	0.29^{***b} (0.09)
MTP coaching intervention			→	Average of all child outcomes	0.03 (0.04)	−0.24^{***a} (0.06)	0.26^{***b} (0.07)
Direct effects, with mediator							
MTP coaching intervention			→	Average of all child outcomes	0.00 (0.04)	−0.19^{*a} (0.09)	0.15$^{\dagger b}$ (0.09)
Indirect effects							
MTP coaching intervention	→	Instructional support	→	Average of all child outcomes	0.03 (0.02)	−0.04^{a} (0.06)	0.17^{*b} (0.07)

Notes. All continuous variables have been standardized to have a mean of zero and standard deviation of one and, thus, all parameter estimates correspond to effect sizes. Estimates in brackets are standard errors. Greater age diversity refers to classrooms with fewer than 42% 4-year-olds, whereas less age diversity refers to classrooms that enrolled roughly 73%+ 4-year-olds. Different superscripts indicate a significant group difference at $p < 0.05$.
$^{\dagger}p < .10$.
$^{*}p < .05$.
$^{**}p < .01$.
$^{***}p < .001$.

(ES = 0.29, $p < 0.001$) than in classrooms that served a larger share of 3- and 5-year-olds (ES = −0.06, ns), and these group differences were statistically significant (Wald $X^2 = 8.44$, $p < 0.01$). Estimates of indirect, or mediated, effects (i.e., path a × path b of Fig. 1) indicated that the MTP coaching intervention resulted in improvements in children's early learning and development in large part because of improvements in teachers' instructional support, but only in classrooms with a greater number of 4-year-olds (ES = 0.17, $p < 0.05$). Put another way, there was evidence of moderated mediation.

Supplementary analyses. As a robustness check, we estimated a series of auxiliary models to ensure that our findings were robust to alternative specifications. First, considering that some teachers also received a coursework intervention during the prior school year (for more details see: Hamre et al., 2012), which was controlled for in the above analyses, we also explored whether there was a coaching × coursework × age composition interaction. Results from these interaction models revealed no further evidence of moderation. Second, in light of some of the literature that has found that classrooms of greater quality attenuate the harmful effects of classroom age diversity (Guo et al., 2014), we also estimated models that considered this possibility with the MTP intervention (i.e., treatment × age composition × classroom quality at baseline interactions). Results from these models also revealed no evidence of moderation as a function of teachers' baseline instructional and emotional support or classroom organization.

Finally, given the varying dosage of the MTP intervention, we also considered whether the dosage of PD mattered. To do so, all control teachers/classrooms were set to zero, and for the intervention teachers, there is variation depending on how many cycles were completed. Results from these analyses revealed that greater intervention fidelity was linked with improvements in instructional support (ES = 0.43, $p < 0.001$), and the threshold was at eight or more cycles. Teachers who participated in fewer than 8 cycles of MTP did not show improvements in instructional support (ES = 0.12, ns), but teachers who participated in eight or more cycles of MTP demonstrated improvements (ES = 0.87, $p < 0.001$). Nonetheless, intervention fidelity was *neither* linked with greater child outcomes nor was there evidence for differential moderation; that is, the age diversity in classrooms reduced the benefits children derived from MTP (and teachers' instructional support) regardless of intervention fidelity.

4. Discussion

The current study evaluated the benefits for children of a coaching intervention that produced improvements in teacher–child interactions in their classrooms (Downer et al., 2014). Participating classrooms were drawn from a wide range of sites across the country that varied in classroom age composition, which we considered as a potential moderator of the effects of a PD coaching intervention for the early learning and development of preschoolers. When taken together, the results of this investigation begin to highlight the potential benefits of a PD coaching intervention for young children and the importance of considering the role of classroom age diversity in this line of research. We discuss the take-home messages of our work below.

In contrast to prior studies of MTP coaching in a single state with University-based coaches that found small effects on 4-year-olds's school readiness (Mashburn et al., 2010), these benefits for children were not replicated in the current study (i.e., path c of Fig. 1), which involved local coaches in multiple preschool sites across the country that served a larger share of the 3-year-old population. This diminished return to MTP coaching when taken to scale mirrors much of the findings in the early childhood literature that has documented similar diminished returns when programs expand from localized communities (Child and Family Research Partnership, 2015). In fact, across all outcomes of interest, the MTP coaching intervention had an aggregate effect size of less than 3% of a standard deviation for direct assessments of 4-year-old's executive functioning and language and literacy development. These findings are nonetheless surprising given the theory of change and existing literature, which suggests that teachers' instructional support and improvements in instructional support (e.g., Allen et al., 2011; Johnson et al., 2016; Mashburn et al., 2008; Weiland et al., 2013) can (and do) result in small improvements in children's school success.

Despite the lack of aggregate impacts of MTP coaching that transmit to preschoolers in these classrooms, our findings highlight potential heterogeneity in these associations as a function of the age composition of classrooms (i.e., the moderating role of path e of Fig. 1). Specifically, 4-year-old children's expressive vocabulary, phonological awareness, print knowledge, and inhibitory control improved when the following conditions were met: (a) children's teachers received the MTP coaching intervention and (b) children were enrolled in classrooms with a higher number of 4-year-olds (i.e., classrooms with less age diversity). Indeed, the benefits of MTP that transmit to children were roughly 25% of a standard deviation greater in classrooms that were less diverse in age (+1 standard deviation, roughly 78% 4-year-olds) than classrooms that were more diverse in age (−1 standard deviation, roughly 33% 4-year-olds), highlighting the importance of considering interventions to improve educational programs in light of the broader conditions present in the classroom. These results also support some of the existing literature that has found that greater age diversity can result in less optimal experiences for children (Ansari et al., 2016; Moller et al., 2008) and places a greater demand on teachers (Ansari & Pianta, 2018). Accordingly, understanding the contextual and systemic factors that potentially influence changes in program impacts is an area of much needed research (Sheridan et al., 2009) as is a focus on training and PD practices that support teachers' ability to individualize instruction and in differentiating skill and age diversity (Connor, Morrison, Fishman, Schatschneider, & Underwood, 2007; Dixon, Yssel, McConnell, & Hardin, 2014).

Although the above benefits did not differ for children who were enrolled in classrooms with a larger number of younger *or* older peers—that is, any source of age diversity equally diminished the benefits of PD—for 4-year-old children's receptive vocabulary, the MTP coaching intervention was more effective in classrooms with a larger number of 5-year-olds. These latter findings mirror some of the peer effects literature that has found that having more skilled, and perhaps older, classmates can be beneficial for children's language development (e.g., Justice et al., 2011; Mashburn et al., 2009). Although speculative, these results might also indicate that, under certain conditions, PD interventions that aid teachers' facilitation of peer-to-peer interactions, and in turn, exposure to more advanced peers, may facilitate children's own receptive language development. It is somewhat surprising, however, that similar patterns did not emerge for children's expressive language skills. One potential explanation is that although exposure to more complex conversation with older peers results in more expansive receptive vocabulary, these outputs develop before children are able to express them (Senechal, 1997). Moreover, other scholars speculate that young children may also underperform on expressive language assessments because of shyness or test anxiety, which may also mask some of the aforementioned associations (Ribeiro, Zachrisson, & Dearing, 2017). Regardless of the underlying reasons for these patterns, what our results make clear is that the extent to which the benefits of PD improves teachers' behavior and transmit to children is in part constrained by characteristics of the classroom, such as age diversity, and the nature of these processes are rather complex and require continued attention in the future.

Beyond heterogeneity in the benefits of MTP for young children, results from our study also revealed one potential explanation as to why the intervention did not result in gains in 4-year-old's early learning and development in classrooms of greater age diversity. Primarily, it is important to emphasize that our results suggest that the MTP coaching intervention was equally beneficial for teachers' instructional support, regardless of the age composition of classrooms (i.e., path a of Fig. 1 was *not* moderated by age diversity). That is, all teachers benefited to a similar degree from the MTP program. Unfortunately, however, the measure used to detect MTP effects on teacher behavior (the CLASS) does *not* capture how teachers differentiate their instruction (see also Burchinal, 2017). Thus, although all teachers improved their instructional support as a result of MTP coaching, whether they improved equally on indicators related to differentiating instruction is unclear. In other words, teachers who teach in classrooms that serve a large number of different age children may have shown less growth in their practices as compared with teachers in classrooms of less age diversity had the outcome of interest actually focused on individualized instruction. The above is of note because our results also reveal that the age composition of classrooms minimized the benefits children derived from high-quality classroom environments (i.e., path b of Fig. 1 *was* moderated by age diversity). Specifically, 4-year-olds exhibited greater gains in early learning and development as a result of classroom quality, but only when there was less age diversity (i.e., there were fewer 3- and 5-year-olds), which mirrors another study done with separate national sample of Head Start children (Purtell & Ansari, 2018).

Thus, the reason why MTP impacts did not transmit to children in all classrooms is that age diversity overshadowed the benefits of higher quality classroom environments. As discussed in a review by Burchinal (2017), most measures of classroom quality in the early childhood field, including the CLASS, are not intended to capture individualized or differentiated instruction. Accordingly, future studies of the MTP intervention (and classroom experiences more generally) should pay closer attention to the experiences of individual children, as it is plausible that at the classroom level children are provided with high-quality interactions, but individual children do not benefit from these interactions unless they are aligned with their specific needs. Relatedly, given the low levels of instructional support in pre-K in general (Burchinal et al., 2016; LoCasale-Crouch et al., 2007) and in this study—even after the MTP intervention—it is also plausible that these low levels of instructional support, although not directly capturing differentiation, are indicative of teachers who are not adequately individualizing their classroom practices.

It is also possible that the MTP coaching intervention may have focused on improving teaching practices that are most ideally suited for 4-year-olds, rather than 2-, 3-, or 5-year-olds. If this were the case, then what this might suggest is that the effects of age diversity in classrooms may partially be attributed to how well-teaching practices focused on instructional support work better for some ages of children than for others, even though mainly 4-year-olds were actually tested. Put another way, all MTP teachers may show a benefit from the coaching sessions in terms of their instructional support, but perhaps the actual instructional practices are most ideally suited to 4-year-olds, and therefore the 4-year-olds benefit more in classrooms where there are more same age children. This alternative explanation could be addressed by examining how well the MTP coaching sessions cover a range of practices that would be appropriate for children ranging from 2 to 5 years of age instead of targeting developmental practices most appropriate for 4-year-olds, which was unfortunately beyond the scope of the data available.

Regardless of the underlying reasons for the lack of MTP impacts for children in age-diverse settings, these aforementioned findings are particularly important in light of emerging studies done in preschool programs that serve children of different ages, which have documented only small associations between classroom quality and children's school success (e.g., Keys et al., 2013; McCoy et al., 2015; Weiland et al., 2013). That is, one potential explanation for the smaller than expected benefits of classroom quality in the existing literature is the age composition of classrooms, which requires continued empirical attention given the rising number of 3-year-olds attending preschool programs across the country. When taken together, what these results suggest is that PD interventions, including the MTP program, should put greater effort in recognizing the challenges that come with teaching in classrooms with greater age diversity. Considering that the majority of schoolteachers feel ill prepared to provide children with differentiated instruction (Manship, Farber, Smith, & Drummond, 2016), one potential target is helping teachers provide children with learning opportunities that match their needs, interests, and mode of learning. An important first step, however, is developing the tools necessary to measure these practices (for further discussion, see: Burchinal, 2017).

Despite these contributions to the early childhood and PD literatures, a few limitations need to be acknowledged. Primarily, although the current study used a randomized control trial to evaluate the impacts of a PD intervention, teachers and their students were not randomly assigned to classrooms as a function of age composition. Next, while a clear strength of our investigation is that teachers reported on the ages of *all* children in their classroom (and not just the study children), we were somewhat limited by the fact that we did not have the exact ages of children, which limits the degree of diversity that exists in our indicator of age composition. Relatedly, factors related to children's age (e.g., level of language skills in the classroom) may be the actual driver of—or partially account for—the documented associations (e.g., Justice et al., 2011; Mashburn et al., 2009). Unfortunately, however, the children sampled as part of the NCRECE Teacher Professional Development Study were largely 4 years of age, which prohibited us from testing this hypothesis. Future studies of classroom age composition, thus, should consider the extent to which age composition captures variation in child skill at the classroom level. We also did not have the data necessary to consider which (or how many) children were the target of the MTP coaching process.

Additionally, even though we documented heterogeneity in treatment effects as a function of age diversity, these associations are somewhat constrained by the lack of overall effects of MTP coaching. But if our results are replicated in the future, then our findings have clear implications for policy and practice. On the one hand, considering that the MTP intervention seems to be more effective in classrooms that serve largely same age children, our findings suggest that the intervention should be modified to assist teachers with the demands of teaching in classrooms of greater age diversity and targeting teachers' ability to provide differentiated instruction. On the other, considering the expansion of preschool programs that seek to enroll younger children, our results might also suggest that enrollment policies for preschool might consider age cutoffs. The age diversity of classrooms, however, are generally not under the control of teachers, but are under the purview of local sites and programs (e.g., funding streams, enrollment numbers, and child–teacher ratios) and, thus, any effort to reduce age diversity would require a systematic effort. However, it is important to note that a higher match in age between study children and their classmates is likely what is most important.

Moreover, although our results demonstrate heterogeneity in treatment effects of a specific PD intervention, these data are not nationally representative, which is important to consider given the potential for diminished benefits of MTP coaching (and other early childhood intervention programs) when taken into the highly diverse preschool settings present at scale (Child and Family Research Partnership, 2015). We also did not investigate the other two CLASS domains (organizational support and emotional support) because the MTP intervention was heavily focused on instructional support. Finally, even though it is certainly true that we adjusted for other classroom compositional factors in our analyses (e.g., percent of children with an IEP, limited English proficiency, etc.), which allowed us to isolate the moderating role of classroom age composition, we only focused on one source of heterogeneity in isolation from other potential factors related to the classroom ecology. This is important to note given recent work that suggests that the breadth and depth of diversity matters for children's early learning and development (Gottfried, 2016). And in light of recent advances in research methods (e.g., person-centered modeling), an important future direction is the consideration of other theoretically relevant factors related to the classroom that might moderate the impacts of PD interventions, both individually and collectively.

With these limitations and future directions in mind, the results of this investigation provide some promising evidence to suggest that a PD coaching intervention targeted at early childhood educators can have downstream benefits for children. MTP has clearly produced improvements in the overall quality of teacher–student interactions during the preschool years (Downer et al., 2014; Pianta, Masburn et al., 2008), but these benefits have not consistently translated into improvements in children's early learning. What our results make clear is that the degree to which children benefit from these interactions is conditioned on the broader classroom ecology, and, in particular, by age diversity, which is necessary to consider in future PD evaluations and in evaluating the impact of preschool programs more generally. When taken together, the presence of moderated mediation of PD benefits on children and teachers is an illustration of this more complex and nuanced model of documenting and understanding program impacts.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ecresq.2018.03.001.

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